



Sunday 8:34 PM



HW1.pdf
PDF

walk me through the setup, i just downloaded the VM image

Yep. Since the assignment explicitly says **all work must occur inside the course VM**, you should redo the OpenClaw/TAMUS setup inside the VM rather than using the Windows installation you made earlier. The required VM is **Ubuntu 24.04 LTS x86-64**, using **NAT networking**, with at least **4 vCPUs and 8 GB RAM**. HW1

Assuming you're on your Windows PC and downloaded the pre-built image from the link in the homework, we'll go in this order:

1. Get the VM running → 2. Configure its hardware/network → 3. Verify Ubuntu → 4. Update Ubuntu → 5. Install Node/OpenClaw → 6. Set up your TAMUS model again → 7. Verify OpenClaw.

1. First, what file did you download?

Look in your Downloads folder. It will probably be something like:

```
ubuntu-24.04-...7z
```

or after extracting it:

```
.vdi  
.vbox  
.ova
```

If it's a `.7z`, **extract it first** using 7-Zip. Don't try to open the `.7z` directly in VirtualBox.

Your homework recommends **VirtualBox 7.2.x on Windows**, with 7.2.14 being their tested baseline. HW1

If you don't already have VirtualBox installed, install VirtualBox 7.2.x.

2. Add the downloaded Ubuntu image to VirtualBox

What you do next depends on what was inside the download.

If you have a `.vbox` file, double-click the `.vbox` file or open VirtualBox and use:

Machine → Add

and select the `.vbox`.

If you have an `.ova` file, use:

File → Import Appliance → choose the `.ova` → Import

If you only have a `.vdi`, create a VM manually:

New → Name: CSCE465-HW1 → Type: Linux → Version: Ubuntu (64-bit)

Then choose the option to **use an existing virtual hard disk** and select the `.vdi`.

Don't start installing OpenClaw yet.

3. Fix the VM settings before starting it

In VirtualBox, select the Ubuntu VM → **Settings**.

Under **System → Processor**, set:

Processors: 4

You can give it more if your PC comfortably supports it, but 4 is enough.

Under **System → Motherboard**, set approximately:

Memory: 8192 MB

That's the assignment's required 8 GB. HW1

Then go to:

Network → Adapter 1

Make absolutely sure it says:

```
Enable Network Adapter: ✓
```

```
Attached to: NAT
```

Do NOT choose Bridged Adapter.

The homework specifically requires NAT and says the VM must not receive an address directly from your home or campus LAN. HW1

So this is good:

```
Attached to: NAT
```

This is **not** good:

```
Attached to: Bridged Adapter
```

4. Boot Ubuntu

Now click **Start**.

Because this is a pre-built VM, Ubuntu should already be installed. You shouldn't need the Ubuntu installer process described in steps 1–3 of the homework.

Log into the pre-built VM using whatever username/password the VM-image download page provided.

Once you're looking at the Ubuntu desktop, open a terminal with:

```
Ctrl + Alt + T
```

From this point forward, **every homework command we're doing should be typed into this Ubuntu terminal**, not PowerShell on Windows.

That's an important difference from what you did previously.

5. Verify the VM before installing anything

Run:

```
Bash
```

```
uname -m
```

You want:

```
x86_64
```

Then:

Bash

```
ip -brief address
```

and:

Bash

```
ip route
```

These are exactly the verification commands required by Task 1.1.

HW1

For `ip route`, you're looking for something resembling:

```
default via 10.0.2.2 dev enp0s3
```

and your VM might have something like:

```
10.0.2.15/24
```

That's typical VirtualBox NAT behavior.

The exact numbers don't have to match those examples.

6. Update Ubuntu

Once those checks look good, run:

Bash

```
sudo apt update
```

Then:

Bash

```
sudo apt full-upgrade -y
```

Then:

Bash

```
sudo apt install -y curl git
```

Those are the exact base-package steps in the assignment.

HW1

If Ubuntu tells you a new kernel was installed, run:

Bash

```
sudo reboot
```

Otherwise you can continue.

7. Install the required Node version

After you're back in Ubuntu, open Terminal again and run:

Bash

```
curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.40.3/install.sh | bash
```

Then:

Bash

```
source "$HOME/.nvm/nvm.sh"
```

Install the homework's required Node version:

Bash

```
nvm install 24.18.0
```

Set it as default:

Bash

```
nvm alias default 24.18.0
```

Verify:

Bash

```
node --version  
npm --version
```

You should see:

```
v24.18.0
```

The assignment specifically calls for Node 24 LTS and tests against **24.18.0**.

HW1

8. Install the course version of OpenClaw

This is another place where your earlier setup matters. **Don't install the newest OpenClaw.**

Run:

Bash

```
npm install -g openclaw@2026.7.1-2
```

Then:

Bash

```
openclaw --version
```

You want it to report:

```
2026.7.1-2
```

The homework explicitly says not to use `openclaw@latest` because the environment needs to be reproducible.

HW1

At this point, **stop**.

You will have:

```
Windows host  
↓  
VirtualBox  
↓
```

```
Ubuntu 24.04 x86-64 VM
├─ NAT networking
├─ 4 vCPUs
├─ 8 GB RAM
├─ Node 24.18.0
└─ OpenClaw 2026.7.1-2
```

Then we'll redo the **TAMUS** `protected.gpt-4o + tamu-shim.mjs` setup inside this VM. That's essentially the setup we got working before, except this time it'll be in the correct isolated environment. The homework requires the shim because OpenClaw's adapter can otherwise hit the HTTP 422 problem you encountered earlier. HW1

For right now, tell me what file(s) you see after extracting the VM image—for example `.ova` , `.vdi` , or `.vbox` —and I'll walk you through the exact VirtualBox clicks from there.

Sources

